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1. Project Overview

WifiDeauthTwin is a wireless security project focused on the simulation and analysis of Wi-Fi Deauthentication and Evil Twin attacks in a controlled environment. The project demonstrates how attackers can disconnect users from a legitimate wireless network and deceive them into connecting to a rogue access point with the same SSID.

The system is implemented using a *NodeMCU ESP8266 Wi-Fi module,* programmed in *Arduino C*, which performs network scanning, deauthentication, rogue access point creation, and phishing page hosting.This project is strictly designed for *educational, ethical, and security-awareness purposes,* highlighting real-world wireless threats and the need for strong Wi-Fi security practices.

2. Project Objectives

- To write Arduino C code for the creation of the WifiDeauthTwin system.
- To scan and identify nearby wireless access points.
- To simulate Wi-Fi deauthentication and Evil Twin attacks in a controlled environment.
- To analyze observed attack behavior and highlight secure wireless configuration and user awareness measures.

3. Technology Stack

Hardware Platform:

- NodeMCU ESP8266 Wi-Fi Module

Programming Language:

- Arduino C

Development Environment:

- Arduino IDE

Networking Technology:

- IEEE 802.11 (Wi-Fi)

Web Interface:

- Embedded Web Server on ESP8266

Attack Techniques (Simulated):

- Wi-Fi Deauthentication
- Evil Twin Attack

4. Core Features

Wi-Fi Network Scanning

- Scans and displays nearby SSIDs
- Allows selection of target Wi-Fi network

Deauthentication Attack

- Disconnects clients from the legitimate router
- Forces victims to search for available networks

Evil Twin Access Point

- Creates a rogue open access point with the same SSID
- Attracts victims to the fake network

Phishing Page

- Displays a fake router firmware update page
- Requests Wi-Fi password from the user

Credential Verification

- Verifies entered passwords in the backend
- Shows error for incorrect credentials
- Accepts correct credentials and proceeds

Credential Logging

- Stores entered passwords
- Displays captured credentials on the control page

5. Technical Implementation

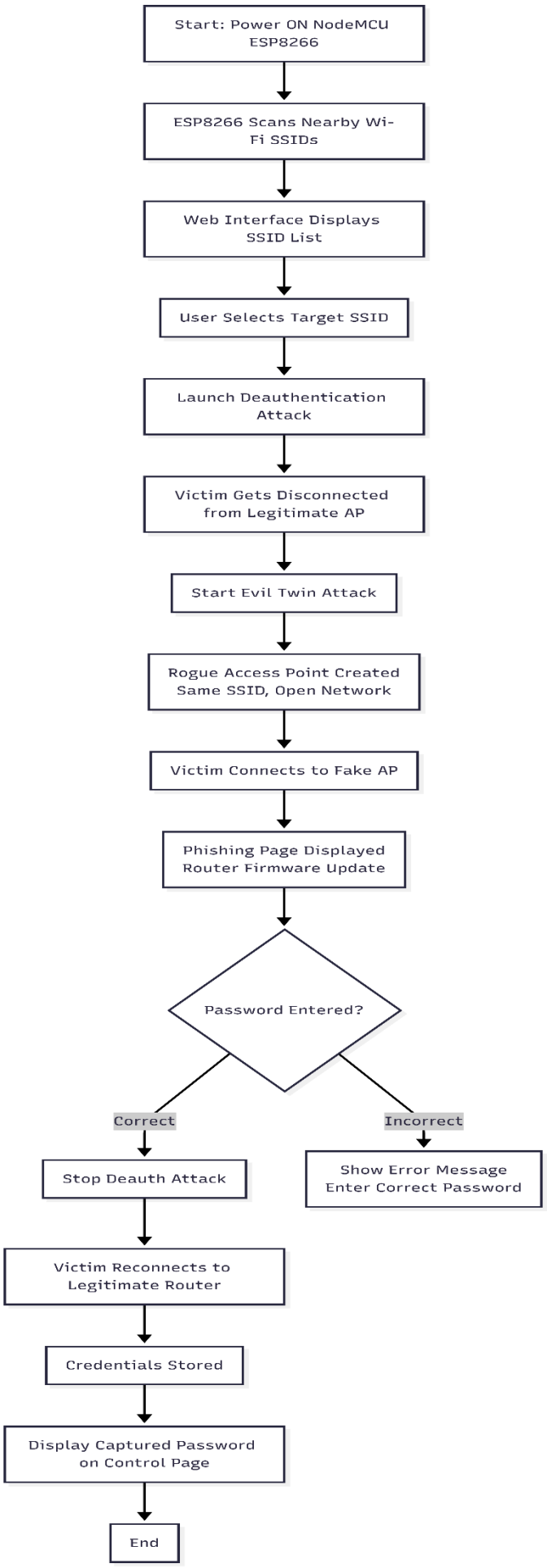
The *WifiDeauthTwin* system uses the ESP8266's Wi-Fi capabilities to scan networks, transmit deauthentication frames, and host a rogue access point with an embedded phishing page.

- Deauthentication packets disconnect legitimate users
- Evil Twin AP mimics the original network
- Phishing page captures and verifies credentials
- Deauthentication is stopped after verification
- Victims reconnect to the legitimate router

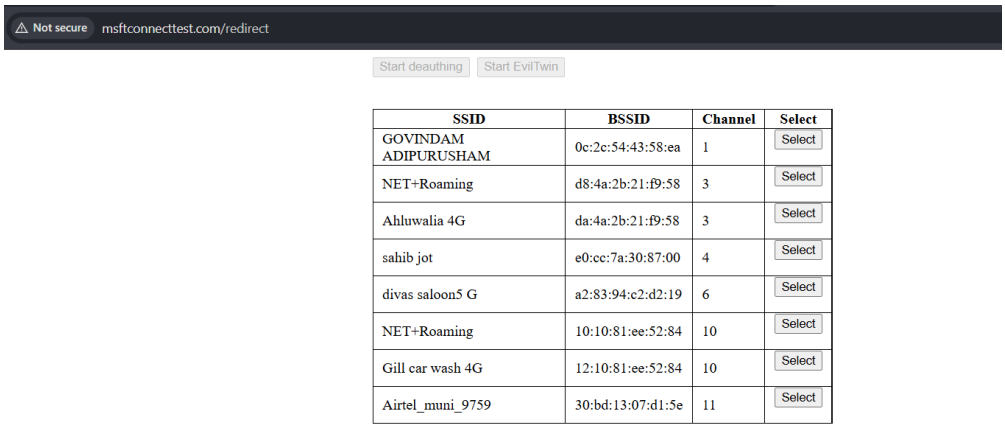
All actions are performed in a **c***ontrolled lab environment*.

6. Application Flow

Following is the flow chart for application flow:

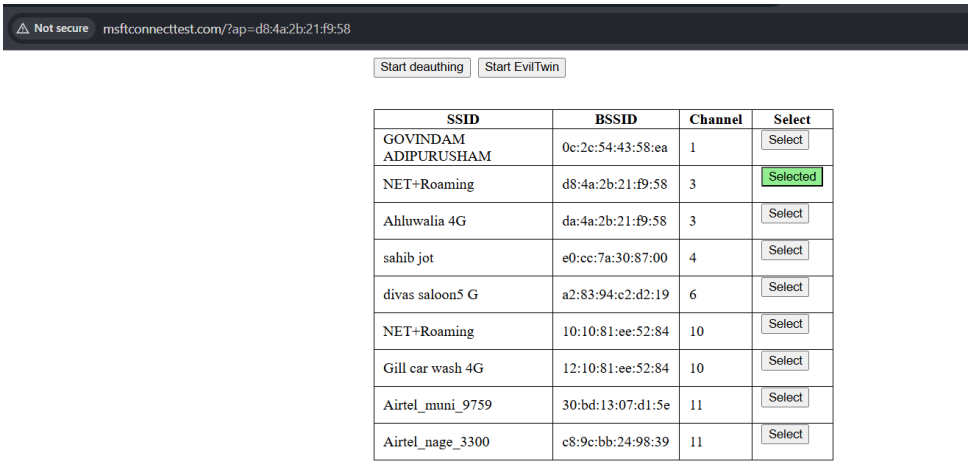

Start: Power ON NodeMCU ESP8266
ESP8266 Scans Nearby Wi-Fi SSIDs
Web Interface Displays SSID List
User Selects Target SSID
Launch Deauthentication Attack
Victim Gets Disconnected from Legitimate AP
Start Evil Twin Attack
Rogue Access Point Created Same SSID, Open Network
Victim Connects to Fake AP
Phishing Page Displayed Router Firmware Update
Password Entered?
Correct
Incorrect
Stop Deauth Attack
Show Error Message Enter Correct Password
Victim Reconnects to Legitimate Router
Credentials Stored
Display Captured Password on Control Page
End

1. ESP8266 scans nearby Wi-Fi networks



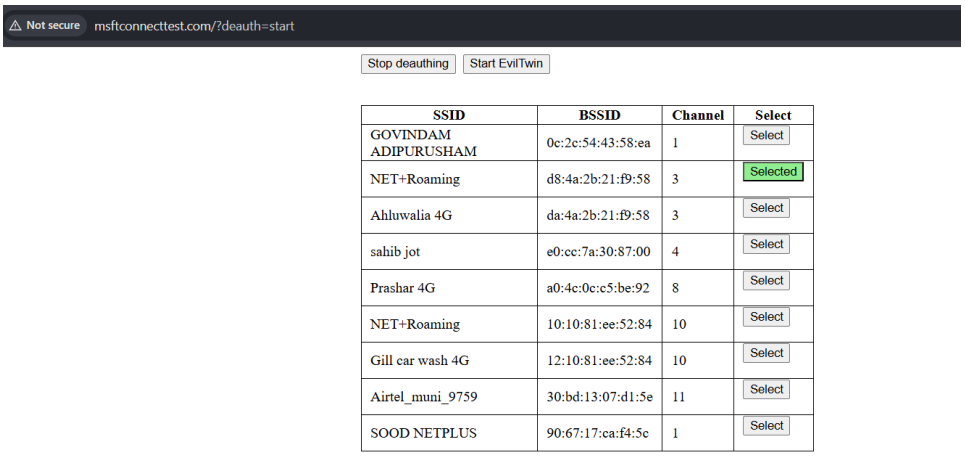
SSID	BSSID	Channel	Select
GOVINDAM ADIPURUSHAM	0c:2c:54:43:58:ea	1	Select
NET+Roaming	d8:4a:2b:21:f9:58	3	Select
Ahluwalia 4G	da:4a:2b:21:f9:58	3	Select
sahib jot	e0:cc:7a:30:87:00	4	Select
divas saloon5 G	a2:83:94:c2:d2:19	6	Select
NET+Roaming	10:10:81:ee:52:84	10	Select
Gill car wash 4G	12:10:81:ee:52:84	10	Select
Airtel_muni_9759	30:bd:13:07:d1:5e	11	Select

2. User selects target SSID



SSID	BSSID	Channel	Select
GOVINDAM ADIPURUSHAM	0c:2c:54:43:58:ea	1	Select
NET+Roaming	d8:4a:2b:21:f9:58	3	Selected
Ahluwalia 4G	da:4a:2b:21:f9:58	3	Select
sahib jot	e0:cc:7a:30:87:00	4	Select
divas saloon5 G	a2:83:94:c2:d2:19	6	Select
NET+Roaming	10:10:81:ee:52:84	10	Select
Gill car wash 4G	12:10:81:ee:52:84	10	Select
Airtel_muni_9759	30:bd:13:07:d1:5e	11	Select
Airtel_nage_3300	c8:9c:bb:24:98:39	11	Select

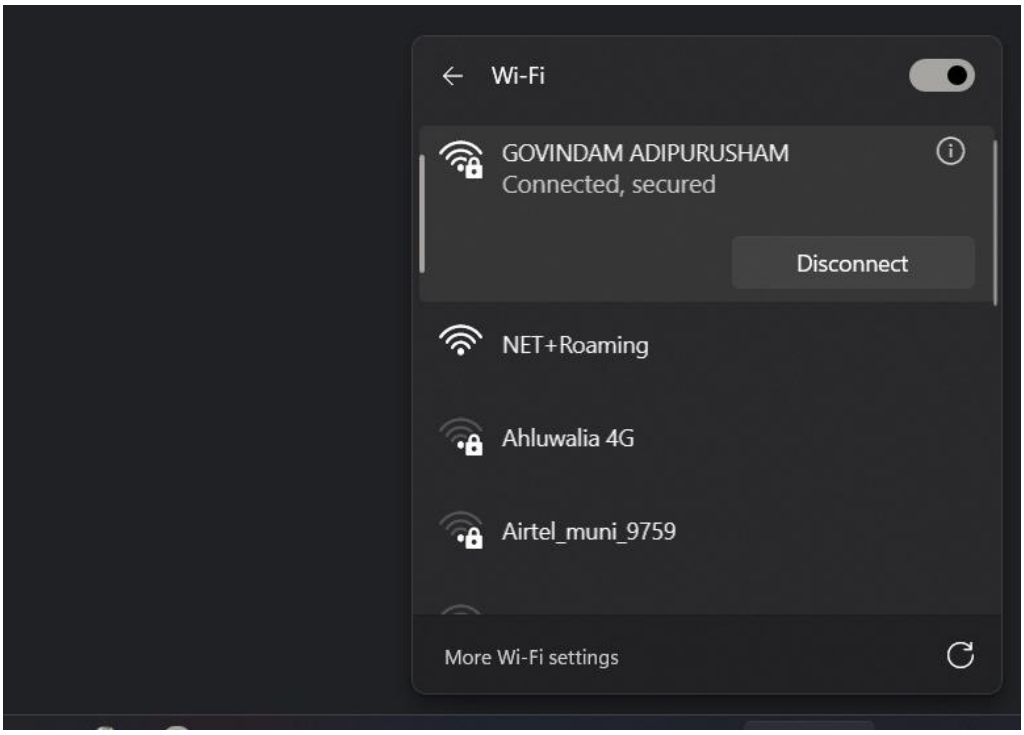
3. Deauthentication attack starts



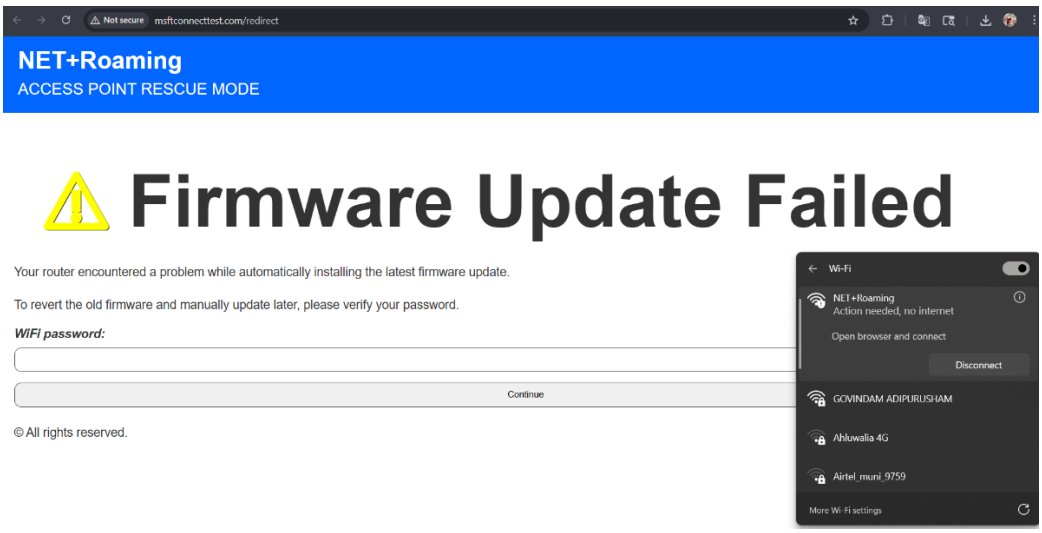
SSID	BSSID	Channel	Select
GOVINDAM ADIPURUSHAM	0c:2c:54:43:58:ea	1	Select
NET+Roaming	d8:4a:2b:21:f9:58	3	Selected
Ahluwalia 4G	da:4a:2b:21:f9:58	3	Select
sahib jot	e0:cc:7a:30:87:00	4	Select
Prashar 4G	a0:4c:0c:c5:be:92	8	Select
NET+Roaming	10:10:81:ee:52:84	10	Select
Gill car wash 4G	12:10:81:ee:52:84	10	Select
Airtel_muni_9759	30:bd:13:07:d1:5e	11	Select
SOOD NETPLUS	90:67:17:ca:f4:5c	1	Select

4. Victim is disconnected

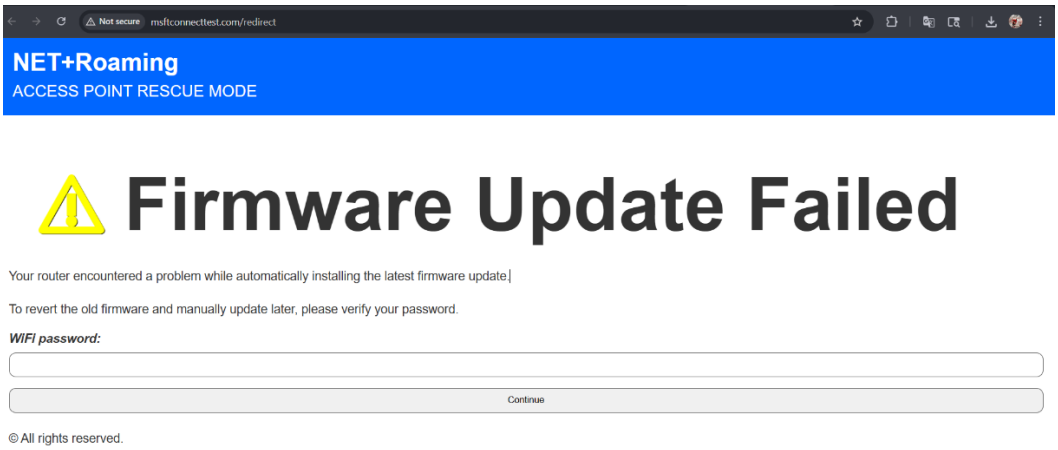
5. Evil Twin access point is created



6. Victim connects to rogue AP

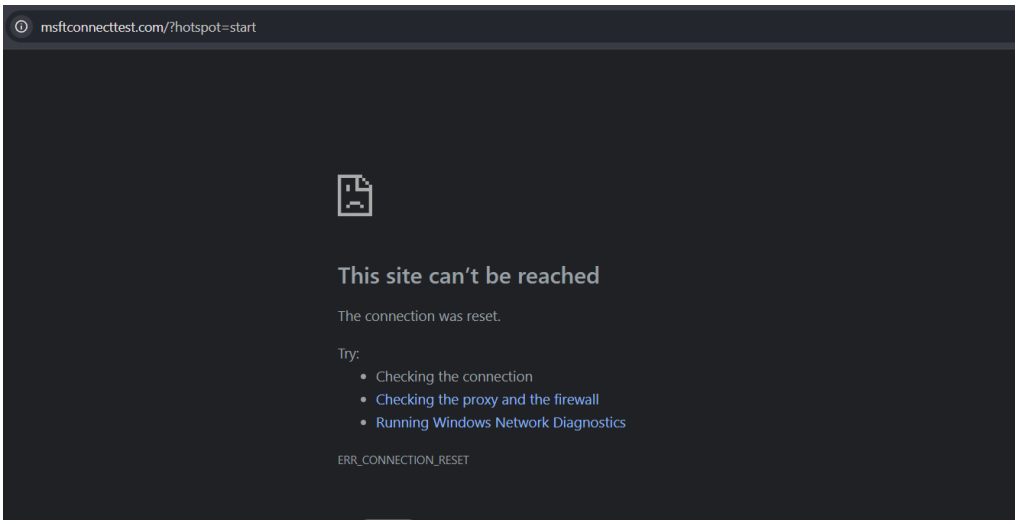


7. Phishing page is displayed



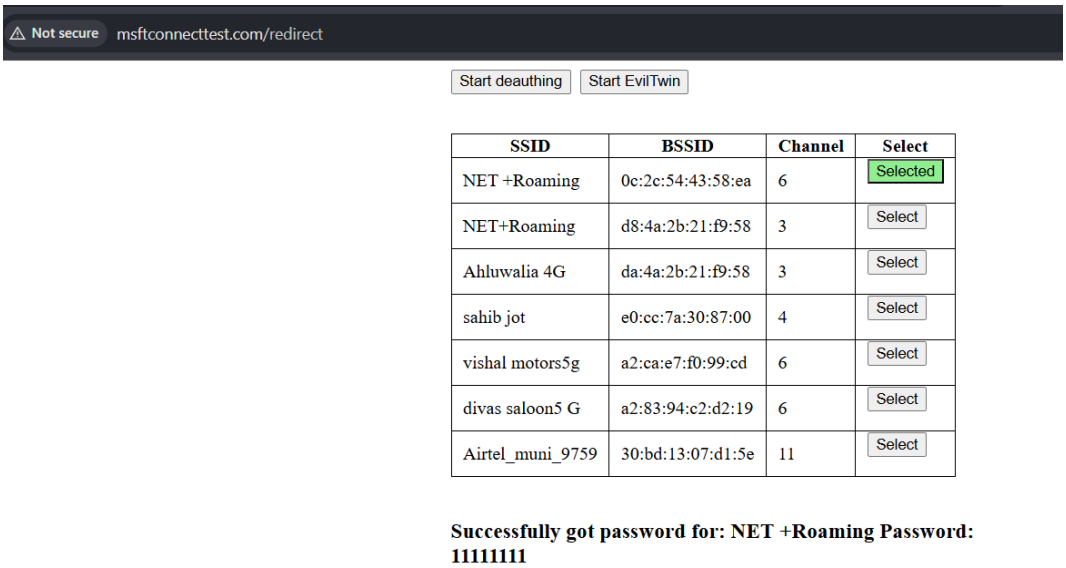
8. Password is entered and verified

9. Deauthentication is stopped



10. Victim reconnects to original router

11. Captured credentials are displayed



7. Technical Challenges and Solutions

Wireless Packet Handling

- **Challenge:** Stable deauthentication transmission
- **Solution:** Optimised packet timing and controlled scope

ESP8266 Resource Constraints

- **Challenge:** Limited memory and processing power
- **Solution:** Lightweight HTML and efficient logic

Credential Verification Handling

- **Challenge:** Managing correct and incorrect inputs
- **Solution:** Backend validation with user feedback

8. Future Enhancements

- WPA3 and 802.11w (PMF) detection support
- HTTPS-based phishing simulation (educational)
- Improved user interface

- Timestamped credential logging
- Evil Twin detection mechanisms

9. Conclusion

The **WifiDeauthTwin** project successfully demonstrates how Wi-Fi deauthentication and Evil Twin attacks can be combined to exploit wireless network vulnerabilities. By simulating these attacks in a controlled environment, the project highlights the importance of secure wireless configurations and user awareness. This project serves as a valuable learning tool for understanding wireless threats and improving network security practices.